



URBAN REDEVELOPMENT AUTHORITY OF PITTSBURGH

LandCare Assurance

Owner Guide

Prepared for the incoming owner, [ura-gis/land-care-assurance](#)

06 August 2026

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Current as of August 6, 2026. For status, sign-off, and outstanding successor work, use [04-readiness-checklist.md](#). This guide explains how the system works and why; the checklist tracks what is done.

1 What you own

LandCare Assurance answers one question for URA supervisors: for this service period, which assigned parcels were actually serviced, by whom, and with what evidence. It is a map-first web application published from this repository. It queries live ArcGIS layers at page load and embeds a secure Power BI report for finance.

Before it existed, contractor compliance was judged from a reported completion figure that blended Active and Request Only parcels against a parcel universe with no ownership check. The application replaces that single percentage with a map, a stated denominator, and a photo per parcel.

1.1 Live routes

All under <https://ura-gis.github.io/land-care-assurance/>.

Route	Audience	Purpose
/monitoring/	Supervisors	Parcel map, evidence, field notes, filters
/kpi/	Leadership	Completion trend, contractor exposure, Power BI finance
/contractor/	Contractors	Progress view and prefilled service submission
/survey-submission/	Legacy links	Redirect to /contractor/
/design-system/example.html	Developers	Portable BI kit reference page

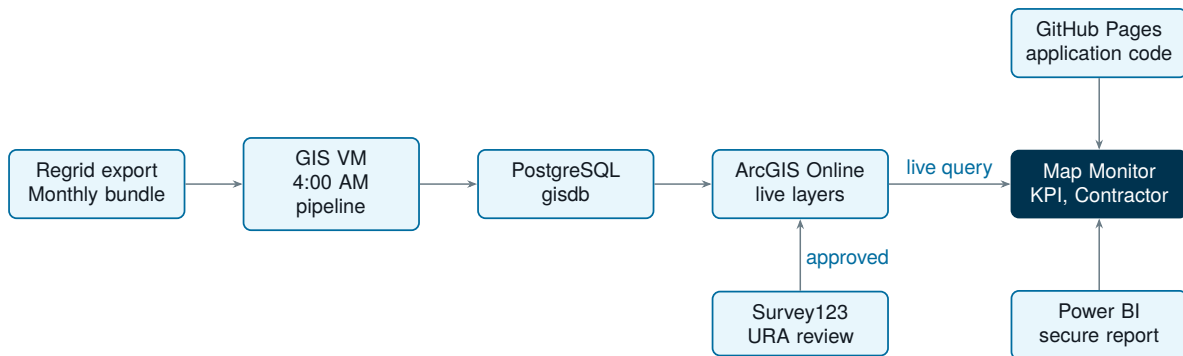
1.2 Service period convention

A LandCare period runs the 15th of one month through the 14th of the next, stored as the 15th of the start month. 2026-06-15 means the June to July period. Every count in the product is scoped to a period. When a number looks wrong, check the selected period first.

2 System architecture and data flow

Three layers with three different owners.

Layer	Where it runs	Owner
Ingestion	GIS VM Task Scheduler at 4:00 AM, repo URA-GIS-User/URA-Data-Repository	GIS and data operations
Store	PostgreSQL gisdb and ArcGIS Online (urap.maps.arcgis.com)	GIS and data operations
Publish and application	This repository, GitHub Pages	Web and dashboard owner
Finance	Power BI semantic model, embedded as a secure report	Finance and BI



The 4:00 AM pipeline is the only scheduled job the product depends on. Everything else is read at page load: ArcGIS supplies assignments, evidence, and completion, GitHub Pages serves the application, and Power BI supplies finance through a secure embed. The former 7:00 AM refresh is deprecated and is not shown.

2.1 The rule that explains most confusion

Live ArcGIS is authoritative at page load. The JSON and GeoJSON committed under docs/landcare/data/ is a compatibility fallback, not the current data source. A number on the map changes during the day with no commit, because the browser queried ArcGIS directly. Do not treat the committed files as truth, and do not report them as freshness.

This became the whole story on 6 August 2026, when the 7:00 AM refresh was deprecated. Nothing on the live pages depends on it now.

2.2 Daily schedule, Eastern time

Time	Task	Output
4:00 AM	regrid_survey_daily_pipeline.py upstream	Regrid CSV into gis.regrid_survey_submissions, then the ArcGIS survey layer
4:15 AM on the 15th	regrid_survey_monthly_export.py	G-drive CSV archive only, not a dashboard source
7:00 AM	refresh_landcare_dashboard.ps1	Deprecated. Retained only as a recovery reference
On demand	ingest_landcare_netsuite_checks.py	Supervised reconciliation only. Not a production feed

The 4:00 AM pipeline is the only scheduled job the product depends on. Everything the supervisor sees is queried live from ArcGIS afterwards, so completion counts rise through the day without any repository publish.

The 7:00 AM job used to publish the static contract under docs/landcare/data/. Its last automatic commit was 28 July 2026 and the checked-in files were last generated on 29 July. The successor's retirement steps are in [04-readiness-checklist.md](#); do not simply delete the task.

2.3 Live ArcGIS sources

All on services1.arcgis.com/0DMNBNaacQNEfN4H.

Layer	Item ID	Used for
gisdb_gis_regrid_surveys	7a2e1d9bacba461296c54a63f104cf51	Evidence, photos, comments, service dates
..._bundle_assignments_current_period	0b4733cb5d204da6ab936c9f6d49e401	Current assignment reference
..._bundle_assignments_history	df7d77eb57f14c68b717c2cf3cdaada4	Monthly assignment denominator
gisdb_gis_epp_parcel_full	live layer	Current URA-owned parcel universe
CouncilDistricts2022	reference	District highlight and filter
LandCare_Survey123_Evidence_Parcels	published layer	Approved evidence, not yet operational

2.4 The adapter boundary

docs/landcare/survey-layer.js and docs/landcare/assignment-layer.js are the only places that talk to ArcGIS. Every source field rename gets absorbed there and nowhere else. That is why additional_comments, additional_notes, additional_note, and notes all normalise to one property, and why service_date, date_of_services, and date_services collapse to one field.

Keep it that way. An ArcGIS URL appearing in monitoring.js or kpi.js is a defect.

2.5 Where to make a change

To change	Open
How survey evidence is read, normalised, matched	docs/landcare/survey-layer.js
How assignments are read and normalised	docs/landcare/assignment-layer.js
Supervisor map behaviour	docs/landcare/monitoring.js
KPI cards, trends, charts	docs/landcare/kpi.js
Contractor progress and submission	docs/landcare/contractor-overview.js
Survey123 URL, prefill names, evidence layer URL	docs/landcare/survey-submission-config.js
Product styling	docs/landcare/app.css
Power BI embed URLs and report pages	docs/kpi/index.html
Native finance card behaviour	docs/landcare/finance-semantic.js
Optional Power BI aggregate extraction	scripts/extract_landcare_powerbi_semantic.py
NetSuite reconciliation import and vendor aliases	scripts/ingest_landcare_netsuite_checks.py
Deprecated 7:00 AM job and its QA gate	scripts/refresh_landcare_dashboard.ps1, scripts/validate_landcare_daily_refresh.py

3 Metrics and the denominator rule

This section is the one to read before changing any number on screen. The full glossary is docs/landcare-metrics-context.md.

3.1 Completion

The numerator is the raw count of survey records whose normalised parcel number matches an assignment in the active filter. The denominator is Active assigned. Request Only parcels

are excluded from the Active denominator.

Repeated records are kept on purpose, so this is **not** a unique-parcel count. Unique completed parcels exist as a diagnostic only. Map Monitor and the KPI dashboard use the same raw matched numerator, which is why their figures agree.

3.2 Definitions in use

Metric	Definition
Current URA-owned LandCare parcels	Live ArcGIS EPP records tagged LandCare with inventory_type = 'URA Owned'
Active assigned	Active assignment records in the selected period and filter
Request Only	Assignment rows at Request Only maintenance level; assigned inventory, not failure
Complete survey records	Raw survey records matched to an assignment parcel in the filter
Unique completed parcels	Distinct assignment parcel keys with at least one match; diagnostic only
Open active	Active assignment parcel keys with no matched evidence; the follow-up queue
Active completion %	Complete survey records over Active assigned; the primary operational KPI
Blended completion %	Complete survey records over assigned total, Request Only included; secondary context
All ArcGIS survey records	Raw survey features for the selected period, including unmatched rows

3.3 Reconciliation rules

Do not compare all ArcGIS survey records directly to complete survey records; the first includes survey-only rows, the second is filtered to assignment matches. Do not compare the current parcel universe to a monthly assignment period; the first is live inventory, the second is a historical export. Treat Request Only as inventory context.

Contract expectations come from the LandCare budgeting workbook, not from ArcGIS. The contract parcel count is contract scope and is a different population from either the EPP universe or a monthly assignment slice. Do not mix them in one ratio.

3.4 Finance actuals

Finance is governed separately from the operational metrics, and it moved twice in three days. Section 5 covers it in full; the rule that matters here is the vocabulary.

Term	What it is
Expected	Workbook contract forecast: term, parcels, monthly and annual amounts
Check requests	Requests posted to the LandCare lawn-maintenance account, governed in Power BI
Other program actuals	Amounts on the same account with no current contractor

A check request is not a cleared payment. It records that payment was requested, not that money moved. Do not describe it as paid, and do not treat a contractor showing zero in a period as proof that no work happened or that no liability exists; it means no matching check request was found in that period.

Quarterly comparisons use only records posted in the selected quarter. Other program actuals are excluded from contractor-to-contract variance, because they are not attributable to a contractor.

4 Daily operations

4.1 The three-minute morning check

1. Open Map Monitor and confirm the latest ArcGIS assignment and survey periods.
2. Compare the selected-period completion count between Map Monitor and KPI. They use the same numerator, so a difference is a defect.
3. Open one Field Note and confirm its parcel, contractor pill, comment, date, and image.
4. Open Land Care Budget and Parcel Area and confirm the secure Power BI report loads.
5. Check the Pages deployment if application code changed.

There is no status JSON to read any more. Freshness lives in the ArcGIS layers themselves and in the Power BI report header. Use `docs/task-scheduler-vm-operations.md` only to archive, disable, or deliberately reactivate the deprecated 7:00 AM process.

4.2 Logs and gates

Artifact	Path	Purpose
Pages validation	<code>.github/workflows/pages.yml</code>	Data contract check on merge; the live gate
QA validator	<code>scripts/validate_landcare_daily_refresh.py</code>	Used by the deprecated job and by manual reconciliation
Daily log	<code>C:\srv\logs\land-care-assurance\daily-refresh-YYYY-MM-DD.log</code>	Historical, from the deprecated job
Status JSON	<code>C:\srv\logs\land-care-assurance\daily-refresh-status.json</code>	Historical; archive it before disabling the task

The Pages workflow does more than deploy. It checks that all routes exist, that the summary JSON, GeoJSON, and refresh manifest agree on counts, and that ownership values stay within URA and Pittsburgh Land Bank. Treat a red Pages workflow as a data contract failure, not a deployment failure.

4.3 Morning executive brief

`.github/workflows/landcare-morning-brief.yml` runs at 9:00 AM Eastern, handling day-light saving by firing at both candidate UTC hours and publishing only in the real window. It sends HTML email through Microsoft 365 Graph when the secrets are present, and otherwise creates an assigned GitHub Issue labelled `landcare-brief`.

After the transfer, set the repository variables `LANDCARE_EMAIL_RECIPIENTS` and `LANDCARE_`

ISSUE_ASSIGNEE, then test with the dry-run delivery mode before going live.

5 Finance: Power BI and NetSuite

Finance moved twice in three days, so read this before touching any money figure.

Date	Change
4 August 2026	NetSuite check-request actuals published into <code>finance_summary.json</code>
5 to 6 August 2026	Land Care Budget and Parcel Area became secure Power BI embeds

5.1 Who owns which number

Power BI is the authoritative reporting source. The Land Care Budget and Parcel Area tabs on the KPI dashboard are authenticated Microsoft embeds, not copied values. Each viewer sees the report through their own Power BI permissions, so the numbers cannot drift from what Finance sees.

NetSuite is the upstream accounting system. It is where check requests originate. The importer in this repository is now a supervised reconciliation tool, not a production feed.

The budgeting workbook still defines contract expectations: term, parcels, and forecast.

5.2 Power BI configuration

Setting	Value
Workspace	GIS Dashboards, A4C26AF1-2334-4FF6-BCCC-FCC7BB0862F5
Semantic model	924c6c0b-6e29-41cf-9775-562ca646953a
Report	2d592a10-7083-470a-96aa-41fbdc59218c
Land Care Budget page	fe756b7016e6baa7351e
Parcel Area Distribution page	4a5502453e9080b7a655
Required filter	LandCare Check Requests[Item Type] = Landcare

Page 8c93bab49c96aa8e3bd2 is Maintenance Check Requests and must never be embedded. It is a different scope and would publish the wrong number. The embed is authenticated, not Publish to web; do not convert it.

Audited on 5 August 2026 the report showed \$458,995.17 spent against a \$775,000 limit, 59.23 per cent, with Q1 \$188,579.00, Q2 \$192,318.50, and Q3 \$78,097.67.

5.3 The honest status of the native finance cards

This is the part most likely to be misdescribed, so state it carefully.

The embedded Power BI tabs are live. The **native** KPI finance cards still read the checked-in `finance_summary.json`, which was generated on 4 August from NetSuite. That file has no `semantic_summary` key, and its `actual_invoice_source.source_system` still reads NetSuite.

Do not describe the native finance feed as Power BI-backed until all three are true:

1. the published JSON contains `semantic_summary`,

2. `actual_invoice_source.source_system` is Power BI semantic model, and
3. the VM status reports `power_bi_finance.feed_status = current`.

Until then the embeds are governed and current, and the native cards are a dated snapshot. `scripts/extract_landcare_powerbi_semantic.py` exists to close that gap and is optional; it is only needed if public KPI finance cards must survive without a Microsoft sign-in.

5.4 Access still outstanding

The audited user can view the shared report but cannot open the workspace. Refresh history, gateway and source connections, and service-principal Build permission all require a workspace administrator. Getting that access is successor work.

5.5 NetSuite reconciliation

Still useful when a Power BI figure looks wrong and you need the underlying requests.

Item	Value
Saved search	All URA LandCare Check Requests, ID 1618
Account	66220 Property Management : Lawn Maintenance
Supporting report	ID 697; funding reference ID 704

Export the saved search to CSV on the secured VM, never into this repository, then run:

```
python scripts\ingest_landcare_netsuite_checks.py `
--source C:\secure\exports\landcare-checks.csv
```

The 4 August read found 628 records totalling \$4,538,233.89; for the cycle from 1 November 2025, 85 records totalling \$618,513.87, of which \$592,179.76 mapped to current contractors and \$26,334.11 stayed as other LandCare program expense.

Do not automate the authenticated Power BI DOM or a browser CSV download to get these numbers. Show as a table and Export data are fine for supervised reconciliation; unattended extraction must use the semantic model Execute Queries API. Browser markup, visual identifiers, and download behaviour are not a stable contract.

5.6 Vendor aliases

NetSuite vendor names do not match contractor names in the assignment layer, so `scripts/ingest_landcare_netsuite_checks.py` holds an alias table. One contractor can appear under several NetSuite names, for example K.R.J. Enterprises Inc and K.R.J. Enterprises Inc. - Eltridra both map to KRJ Enterprises.

An unrecognised vendor is not dropped. It stays in `other_program_actuals`. **Add an alias only after Finance or the LandCare program owner confirms the relationship.** Guessing moves money onto the wrong contractor's variance line.

5.7 What is published and what is not

finance_summary.json carries aggregates only: one row per posting month and current-cycle contractor, monthly other-program amounts, and source metadata. Document numbers, memos, and transaction-level vendor records are deliberately excluded, because GitHub Pages serves this file to anyone.

scripts/validate_landcare_daily_refresh.py fails the publish if the published contractor aggregates do not sum to current_cycle_contractor_total. If that fails, re-run the import rather than editing the JSON.

5.8 Access and security

Use named accounts with read-only report access for both systems. Never store passwords, session cookies, browser profiles, certificates, raw exports, document numbers, or memos in GitHub. The Power BI certificate lives only under C:\srv\secrets\powerbi\.

Full detail in docs/powerbi-landcare-finance-source.md, docs/powerbi-landcare-dataflow-audit-2026-08-05.md, and docs/netsuite-landcare-finance-source.md.

6 Replacing Regrid with a URA front end

6.1 Scope it honestly

Regrid provides one thing URA does not yet own: the contractor capture form and its daily export, which lands in gis.regrid_survey_submissions and publishes to gisdb_gis_regrid_surveys. Assignments, parcel geometry, ownership, the dashboard, and the contractor portal are already URA property.

You are replacing an intake form and an export, not a platform. ArcGIS stays.

6.2 Most of it is already written

Piece	Where	State
Contractor portal, map and list parcel selection	docs/contractor/, contractor-overview.js	Live
Survey123 intake and prefill contract	survey-submission-config.js	Live
Webhook receiver	landcare_survey123_webhook.py	Written, not deployed
Evidence validation and sync	survey123_evidence_sync.py	Written, not deployed
Evidence layer publisher, REST, no ArcGIS Pro licence	publish_landcare_survey_evidence_parcel.py	Written, not deployed
Database migration	sql/20260728_landcare_survey123_evidence_parcel.sql	Not applied

6.3 What is left

Apply the migration. Set the VM environment variables listed in docs/survey123-landcare-network-setup.md. Run the receiver behind the approved URA HTTPS host. Register the Survey123 webhook for new records and edits with the X-LandCare-Webhook-Token header. Bootstrap the evidence hosted layer once in the LandCare - Published Layers folder and

record its item ID. Then run in parallel with Regrid.

6.4 Three gates

Stage	Outcome	Done when
1. Intake	Contractor picks a parcel on the URA map or list, Survey123 opens prefilled with organisation, parcel, address, and period	Map tap and dropdown produce the same prefill for the same OBJECTID; an anonymous browser loads the form without sign-in
2. Review and evidence	URA approves or rejects in a restricted view; only approved records reach the public evidence layer	One approved submission creates exactly one database row and one evidence feature, even after a webhook retry; pending and rejected records appear nowhere public
3. Cutover	Map Monitor and KPI read the URA evidence source instead of Regrid	Two full service periods of parallel running with matching counts, parcel matches, contractors, dates, comments, and photos

6.5 Guardrails, not preferences

Never publish the raw Survey123 point layer or the review QA view on a public map. No publisher token, password, or portal credential goes into GitHub Pages, because Pages content is public. Public views are add-only for submission and query-only for reference. Only approved evidence becomes visible.

6.6 After sign-off

Repoint Map Monitor and KPI, update docs/landcare-metrics-context.md and the tests in the same pull request, then retire the Regrid scheduled task and its credential. Retiring Regrid also removes the browser-automation download step, which is the most fragile part of the current pipeline.

If Survey123 later constrains the contractor experience, replace only the form. Build a custom LandCare form that posts to a protected URA API which validates and writes to PostgreSQL or a secured feature service. Keep the browser adapter contract unchanged: parcel number, period, contractor, service date, image URL, and canonical additional_notes. The application does not need to know which form produced the record.

7 Working with AI agents

This repository was built with agent-assisted development and is set up to keep working that way. The detailed prompt library is handover/02-agent-playbook.md. This section is the summary.

AGENTS.md at the repository root is read automatically by Codex and by Claude Code. It carries the source-of-truth rules, the change rules, and the verification requirement. Keep it current. It is the cheapest way to stop an agent from doing the wrong thing.

Every prompt should carry four blocks: the files to read first, the contract (goal, source, metric rule), the commands the agent must actually run, and a task-specific snippet.

7.1 Four failure modes specific to this repository

1. **Denominator drift.** An agent will switch the numerator to unique parcels because it looks more correct. It is not the agreed metric and it changes every published number.
2. **Stale cache-busting.** Modules import each other with `?v=` strings, for example `survey-layer.js?v=20260803-raw-completion-v2`. Change a shared adapter and every importer's version must move together, or browsers serve a mixed bundle.
3. **Hardcoded endpoints.** ArcGIS URLs belong in the two adapter files and in `survey-submission-config.js`, nowhere else.
4. **Hand-editing generated data.** Files under `docs/landcare/data/` come from the VM refresh, so hand edits are silently reverted by the next run.

7.2 Verification before any push

```
python -m unittest discover -s tests
node --test tests/survey-layer.test.mjs
```

`tests/test_handover_contract.py` guards the handover file set and the portability of the design system CSS. Then open the four routes and confirm counts, comments, photos, field notes, filters, and the selected period.

8 Design system

There are three style sources and they are not interchangeable. Knowing which is which saves a great deal of confusion.

Source	What it is	Use it for
<code>docs/landcare/app.css</code>	About 2,800 lines, the live product style, coupled to the ArcGIS JS API 4.30 theme	Changing an existing page
<code>.claude/skills/ura-landcare-design/</code>	The formalised design system extracted from <code>app.css</code> : tokens, 8 components, guidelines, 3 screen templates	Designing anything new, with or without an agent
<code>docs/design-system/executive-bi-css</code>	156 portable lines with its own scoped <code>--bi-*</code> tokens, same palette, different naming	A standalone BI dashboard outside LandCare

The skill's `tokens/colors.css` is an exact extraction of the `:root` block in `app.css`, so the product and the design system cannot drift apart silently. `github.md` inside the skill carries a screen map from each design artifact back to its repo source.

The export was taken before the NetSuite deployment, so its KPI template and some copy examples still show the pre-NetSuite wording such as "NetSuite feed required". Tokens, components, and visual rules are unaffected. See `.claude/skills/ura-landcare-design/SYNC-NOTES.md`.

8.1 Invoking it

The design system is installed as a Claude Code skill. Any agent opening this repository discovers it automatically and can be asked to use it by name:

```
/ura-landcare-design
```

It also works as plain reference material. Open `readme.md` inside the skill folder for the full content and visual rules.

8.2 The rules that get checked in review

One brand blue family carries almost everything. Orange means risk or action needed, exclusively. Green means success, target, or complete, exclusively. Gold is a rare tertiary accent. Never more than these four hues plus ink, muted, and line neutrals on a page.

Manrope only, weights 400 to 800, no serif or mono in the UI. Headings are heavy with tight negative letter-spacing; body copy is never bold. A small uppercase kicker precedes every heading.

Flat backgrounds, no photography, no texture, and exactly one gradient reserved for the single featured metric card per band. Hairline 1px borders, two shadow tokens, corner radii of 12, 18, 22, and full pill. Animation limited to short colour transitions on hover.

Never use colour alone; pair every status colour with a text label. Every important number shows its source, period, denominator, and freshness. Normal text contrast at least 4.5:1. Tabs and actions are real buttons or links with `aria-selected`. Tables keep `table`, `th`, and `td`. Layout stacks at 760px and below. Loading, empty, stale, and error states are explicit, because live ArcGIS queries fail in ways static pages do not.

Copy is operational and procedural, never marketing-toned. Sentence case throughout. No emoji, no exclamation points, no filler adjectives. State unavailable data directly rather than showing a vague empty state.

One caution: `tokens/typography.css` imports Manrope from Google Fonts, so the skill's templates need network access. This does not affect `executive-bi.css`, which is deliberately import-free and is asserted so by `tests/test_handover_contract.py`.

9 Ownership, risks, and open decisions

Five areas have distinct owners: ArcGIS source freshness, GitHub Pages and application code, the Power BI report, contractor follow-up, and retiring the deprecated 7:00 AM task. The escalation table lives in `HANDOVER.md`.

9.1 Open items to decide, not defects

Item	What it means for you
Survey123 evidence path is code-complete but not deployed	Approved photos cannot appear and Regrid stays the only evidence source. The single blocker on the replacement

Item	What it means for you
Native KPI finance cards are not yet Power BI-backed	The embeds are governed and live, but the native cards read a 4 August NetSuite snapshot. Decide whether public cards need to survive without a Microsoft sign-in
Power BI workspace administration is not granted	Refresh history, gateway connections, and service-principal Build permission all need a workspace administrator
The deprecated 7:00 AM task is still on the VM	Archive its definition and logs, disable it, then remove after two clean business days
One check request, \$26,334.11, sits in other program actuals	It is on the LandCare account with no current contractor. Ask Finance whether it should be aliased to a contractor or stay separate
Survey123 submissions do not change official completion in v1	Whether they should is a governance decision, not a code change
Checked-in data was last generated 29 July 2026 while ArcGIS moves daily	Static and live numbers differ. Decide whether the compatibility fallback stays at all
Morning brief falls back to a GitHub Issue without Microsoft 365 secrets	Set the two repository variables and test with dry-run first
GitHub redirects the old repository path, Pages URLs do not	Every ArcGIS embed, bookmark, and operational link moves to the ura-gis Pages URL by hand

9.2 First week

1. Confirm you can clone, push, and run the tests.
2. Watch two 4:00 AM ArcGIS publication cycles and confirm the layer timestamps move.
3. Run the morning brief manually in dry-run mode and read the artifact.
4. Open all four routes and reconcile one contractor's numbers by hand against the map.
5. Open Land Care Budget and Parcel Area and confirm the authenticated report loads for you.
6. Request Power BI workspace administration access; several open items depend on it.
7. Archive and disable the deprecated 7:00 AM task, following the readiness checklist.
8. Make one harmless documentation change through an agent, end to end, including tests.
9. Decide whether to deploy the Survey123 evidence path this quarter.

10 Appendix

10.1 Repository map

```

land-care-assurance/
  handover/          <- start here; 04-readiness-checklist.md is the status page
  AGENTS.md          <- agent start rules, read automatically
  HANDOVER.md        <- cutover checklist
  .claude/skills/ura-landcare-design/ <- design system skill
  docs/
    monitoring/ kpi/ contractor/      <- the three app routes
    landcare/          <- app JS, CSS, assets, and published data contract
    design-system/     <- portable Executive BI kit
    *.md               <- architecture, metrics, runbooks
  scripts/            <- VM daily refresh, data build, brief generation

```

```
sql/           <- Survey123 evidence migration
tests/         <- Python and Node tests
.github/workflows/ <- Pages deploy and morning brief
```

10.2 Published data contract

Checked-in under docs/landcare/data/. Last generated 29 July 2026 by the now-deprecated 7:00 AM job, and kept only for compatibility.

File	Contents
all_months.geojson	URA-owned assignment rows with geometry
latest_month.geojson, latest_month_summary.json	Latest month slice
monthly_metrics.json	Completion rates by month
contractor_monthly.json	Contractor completion by month
kpi_summary.json	Summary metadata and latest metrics
finance_summary.json	Workbook and NetSuite aggregates; compatibility only
refresh_manifest.json	Freshness, counts, survey metadata

10.3 Environment variable names

Names only. Values are VM-local or repository secrets and are never committed.

VM .env: PG_HOST, PG_PORT, PG_DB, PG_USER, PG_PWD.

Survey123 evidence path: LANDCARE_PG_DSN, LANDCARE_SURVEY_WEBHOOK_TOKEN, SURVEY123_FEATURE_LAYER_URL, SURVEY123_ARCGIS_TOKEN, SURVEY123_PUBLIC_ATTACHMENT_LAYER_URL, LANDCARE_ASSIGNMENT_HISTORY_LAYER_URL, LANDCARE_SURVEY_EVIDENCE_ENABLED, LANDCARE_SURVEY_EVIDENCE_AGOL_ITEM_ID, LANDCARE_SURVEY_EVIDENCE_LAYER_URL, LANDCARE_SURVEY_EVIDENCE_ARCGIS_TOKEN.

Power BI extraction, only if the optional aggregate path is enabled: LANDCARE_POWERBI_TENANT_ID, LANDCARE_POWERBI_CLIENT_ID, LANDCARE_POWERBI_CERTIFICATE_PATH, LANDCARE_POWERBI_CERTIFICATE_THUMBPRINT, LANDCARE_POWERBI_WORKSPACE_ID, LANDCARE_POWERBI_DATASET_ID. The certificate lives only under C:\srv\secrets\powerbi\. Removing these four authentication variables rolls the extraction back without losing the published contract.

NetSuite reconciliation: LANDCARE_NETSUITE_CHECKS_CSV, the path to the exported CSV.

GitHub Actions secrets: M365_TENANT_ID, M365_CLIENT_ID, M365_CLIENT_SECRET, M365_SENDER_UPN. Repository variables: LANDCARE_EMAIL_RECIPIENTS, LANDCARE_ISSUE_ASSIGNEE.

10.4 Command reference

```
python -m unittest discover -s tests           # Python tests
node --test tests/survey-layer.test.mjs       # survey adapter tests
python handover/build/build_report.py         # rebuild this PDF
```

On the VM:

```
.\scripts\refresh_landcare_dashboard.ps1 `
-RepoRoot C:\srv\GISWebApp\land-care-assurance
.\.venv\Scripts\python.exe scripts\validate_landcare_daily_refresh.py
```

10.5 Further reading

- docs/landcare-architecture.md for source and runtime architecture
- docs/landcare-metrics-context.md for the full metric glossary
- docs/powerbi-landcare-finance-source.md for the Power BI model, embed pages, and rollback
- docs/powerbi-landcare-dataflow-audit-2026-08-05.md for the audit evidence
- docs/netsuite-landcare-finance-source.md for the NetSuite saved search and reconciliation
- docs/landcare-submission-and-evidence-flow.md for the contractor intake contract
- docs/survey123-landcare-network-setup.md for Survey123 and webhook configuration
- docs/task-scheduler-vm-operations.md for the VM runbook and failure triage
- docs/github-handover-runbook.md for the repository and Pages cutover